SIMATS ENGINEERING
ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 5

Code of Conduct
I S. Jefna Princy- 192572058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ___________________

Assessment Tasks
Reverse Engineering Task
For each task, study the described big data solution, infer how it was built, identify the underlying components, and explain what design decisions were likely made.

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy,

encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.

4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.
5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.

Reverse Engineering Task Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Inference Accuracy	25%	Infers system components and flow with high accuracy	Mostly accurate inference	Basic inference	Weak inference
Understanding of Architecture	25%	Shows clear understanding of underlying big data architecture	Good understanding	Partial understanding	Poor understanding
Use of Technical Evidence	20%	Uses strong reasoning and technical evidence	Reasonable evidence	Limited evidence	No meaningful evidence
Depth of Analysis	15%	Analysis is deep and insightful	Reasonably deep	Basic depth	Superficial analysis
Clarity	15%	Clear and organized explanation	Generally clear	Somewhat unclear	Poorly presented

REVERSE ENGINEERING TASK

1. Big Data Platform for Website Clicks and Customer Transactions

The described platform handles billions of website clicks, mobile interactions, and customer transactions every day. From this scale, the system can be inferred to use a **distributed big data architecture** capable of storing and processing both structured and semi-structured data.

Probable Architecture

The storage layer is likely based on a **distributed file system or cloud object storage**, where structured transaction records and semi-structured clickstream data are stored separately before analysis. An ingestion framework such as **Apache Kafka** can collect continuous events, while batch-processing tools such as **Apache Spark** can process large historical datasets.

For real-time analytics, stream-processing engines such as **Spark Streaming or Apache Flink** are likely used. Data can be partitioned based on attributes such as customer ID, region, or timestamp to distribute workloads across multiple nodes.

Analysis

The architecture supports scalability by adding processing and storage nodes as the workload increases. Structured data can be indexed for faster queries, while semi-structured clickstream data can be processed using distributed computing frameworks.

The processed information is finally connected to **dashboards and Business Intelligence tools**, allowing organizations to monitor website activity, customer behavior, transactions, and business performance.

Inferred flow:

Website / Mobile Apps / Transaction Systems

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Data Ingestion Layer

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Distributed Storage

↓

Batch + Stream Processing

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Analytics Layer

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Dashboards / Business Intelligence

2. E-Commerce Platform for Real-Time Personalization

The e-commerce platform collects cart additions, abandoned checkouts, product views, and payment behavior from different regions. This indicates an event-driven architecture designed to process customer activities

continuously and provide personalized recommendations.

Probable Architecture

Event-streaming tools such as Apache Kafka can collect clickstream and transaction events in real time. Customer profiles and transaction records may be stored in distributed databases, while caching systems such as Redis can provide fast access to frequently requested product and customer information.

The data pipeline likely combines structured order data with semi-structured browsing events. Apache Spark or similar distributed processing frameworks can perform large-scale analytics and machine learning operations.

Analysis

Real-time synchronization allows customer activities to immediately update profiles and recommendation models.

Distributed processing improves scalability because workloads can be divided among multiple nodes.

Caching reduces response time for frequently accessed information, while fault-tolerant distributed systems help maintain service availability even when individual components fail.

The resulting recommendations and KPIs can be displayed through dashboards for monitoring sales, abandoned carts, customer engagement, and regional performance.

Inferred flow:

Website / Mobile Users

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Event Collection

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Event Streaming System

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Processing + Customer Profile Store

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Recommendation Engine

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Personalized Results

3. Big Data Healthcare System

The healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Since these include both structured and unstructured information, the system is likely based on a hybrid data architecture.

Probable Architecture

Structured patient and prescription records can be stored in relational databases, while diagnostic images and other unstructured information can be stored in distributed or cloud storage. Data integration tools can connect hospitals, laboratories, insurance systems, and other healthcare sources.

Real-time wearable sensor data can be processed using streaming technologies such as Kafka and Spark Streaming. Machine learning frameworks can then analyse patient information to support disease prediction and continuous monitoring.

Analysis

Healthcare data requires strong privacy and security controls. Encryption, authentication, access control, and auditing are likely used to protect sensitive patient information and support compliance requirements.

The analytics pipeline can combine patient history, diagnostic information, prescriptions, and sensor readings to identify patterns useful for clinical decision-making and population health studies.

Inferred flow:

Hospitals / Laboratories / Wearable Devices
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Data Integration Layer
↓
Hybrid Data Storage
↓
Stream + Batch Processing
↓
Machine Learning / Analytics
↓
Clinical Decision Support

4. Smart City Big Data Platform

The smart city platform continuously collects traffic sensor data, CCTV metadata, weather updates, and public transport activity. The continuous nature of these sources suggests an edge and streaming-based architecture.

Probable Architecture

Edge computing devices can collect and perform initial processing of sensor information close to its source. A distributed messaging system such as Kafka can transfer continuous data to the central platform.

Time-series databases can store traffic and sensor measurements, while location-aware databases can support geographical queries. Streaming analytics can then identify congestion patterns and predict traffic conditions.

Analysis

Batch processing can be used for historical traffic analysis, while real-time processing provides immediate information about current congestion. Combining both approaches helps urban planners understand long-term trends as well as current conditions.

Security can be implemented using authentication, authorization, encrypted communication, and access-control mechanisms. Visualization systems and dashboards can present live traffic conditions and infrastructure information to city authorities.

Sensors / CCTV / Weather / Transport Systems
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Edge Computing Layer
↓
Messaging / Data Ingestion

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Distributed Storage

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Stream + Batch Analytics

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Urban Planning Dashboard

5. Healthcare Analytics and Predictive Health System

The healthcare analytics system integrates patient records, wearable readings, laboratory reports, and appointment histories. The combination of structured medical records and unstructured information indicates a hybrid storage and processing architecture.

Probable Architecture

ETL pipelines can extract data from hospitals, laboratories, wearable devices, and other healthcare systems, transform it into compatible formats, and load it into a centralized analytics environment. Interoperability standards can help different healthcare systems exchange information.

Distributed processing frameworks can analyse large volumes of patient and wearable data. Machine learning models can identify patterns related to disease risk and support predictive health analytics.

Analysis

Because healthcare information is sensitive, the platform requires encryption, authentication, role-based access control, and auditing. Data governance mechanisms can control who is allowed to access different categories of medical information.

Real-time wearable readings can support continuous monitoring, while historical medical and laboratory data can be used for predictive analysis. Machine learning can therefore assist in disease-risk scoring and treatment recommendations, while healthcare professionals remain responsible for clinical decisions.

Inferred flow:

Patient Records + Wearables + Labs + Appointments

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ETL / Data Integration

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Hybrid Storage

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Distributed Processing

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Machine Learning

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Risk Scoring + Treatment Recommendations